Permutations and Combinations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Permutations & Combinations Practice Exercises**

Find the number of distinct arrangements of the letters of the word COMMITTEE.

Final Answer: 45360. Applying Permutations & Combinations principles, $\frac{9!}{2!\,2!\,2!} = 45360$ Therefore, the verified correct option is 45360.

2 **[Foundational]** • **Target: Fundamental Counting and Linear Permutations**

A group of n students, which includes Ali and Priya, are to be arranged in a straight line for a photograph. The number of different arrangements in which Ali and Priya are not standing next to each other is 3600. Find the value of the integer n.

Final Answer: 7. We can find the number of arrangements where Ali and Priya do not stand next to each other using the complementary method or the insertion (gap) method. **Method 1: Complementary Counting** 1. The total number of unrestricted arrangements of the n students in a straight line is $n!$. 2. Consider the arrangements in which Ali and Priya *do* stand next to each other: - Treat Ali and Priya as a single block. There are now $(n-1)$ entities to arrange, which can be done in $(n-1)!$ ways. - Within.

3 **[Foundational]** • **Target: Permutations with Repetitions and Identical Items**

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

Final Answer: 13. To form a committee of 5 members: 1. **Committees with exactly 3 men:** Since there are 5 members in total, the committee must contain 3 men and $5-3=2$ women.
Number of ways $= \binom{n}{3} \times \binom{6}{2} = \binom{n}{3} \times \frac{6\times5}{2} = 15\binom{n}{3}$ 2. **Committees with exactly 4 men:** The committee must contain 4 men and $5-4=1$ woman. Number of ways $= \binom{n}{4} \times \binom{6}{1} = 6\binom{n}{4}$ 3. **Equating the two expressions:**.

© ClipSAT — clipsat.org | All Rights Reserved

4 [Foundational] • Target: Combinations and Committee Selections

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed? Show your complete algebraic work.

Final Answer: 10. The problem requires us to find the number of combinations of selecting 3 people from a group of 5. This is a direct application of the combination formula ${}^{n}C_{r}$, where n is the total number of items to choose from, and r is the number of items to choose. In this case, $n = 5$ and $r = 3$. Using the formula: ${}^{5}C_{3} = \frac{5!}{3!\,(5-3)!} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times (2 \times 1)} = \frac{5 \times 4}{2} = \frac{20}{2} = 10$. Therefore, there are 10 different committees.

5 [Foundational] • Target: Restricted Arrangements and Separations

Find the number of distinct arrangements of the letters of the word SUCCESSFULLY.

Final Answer: 19958400. Applying Permutations & Combinations principles, $\frac{12!}{3!\,2!\,2!\,2!} = 19958400$ Therefore, the verified correct option is 19958400.

6 [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word ENGINEERING.

Final Answer: 1663200. Applying Permutations & Combinations principles, $\frac{11!}{3!\,3!\,2!\,2!} = 1663200$ Therefore, the verified correct option is 1663200.

7 [Foundational] • Target: Fundamental Counting and Linear Permutations

Find the number of distinct arrangements of the letters of the word ARRANGEMENT.

Final Answer: 4989600. Applying Permutations & Combinations principles, $\frac{11!}{2!\,2!\,2!\,2!} = 4989600$ Therefore, the verified correct option is 4989600.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Permutations with Repetitions and Identical Items

Find the number of distinct arrangements of the letters of the word MATHEMATICS.

Final Answer: 9979200. Applying Permutations & Combinations principles, $\frac{11!}{2!\,2!\,2!} = 9979200$ Therefore, the verified correct option is 9979200.

9 [Intermediate] • Target: Combinations and Committee Selections

Find the number of distinct arrangements of the letters of the word ACCOUNTANT.

Final Answer: 302400. Applying Permutations & Combinations principles, $\frac{10!}{2!\,2!\,2!\,2!} = 302400$ Therefore, the verified correct option is 302400.

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Restricted Arrangements and Separations

In how many distinct ways can 12 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 39916800. Applying Permutations & Combinations principles, $(n-1)! = (12-1)! = 39916800$ Therefore, the verified correct option is 39916800.

11 [Intermediate] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word MISSISSIPPI.

Final Answer: 34650. Applying Permutations & Combinations principles, $\frac{11!}{4!\,4!\,2!} = 34650$ Therefore, the verified correct option is 34650.

12 [Intermediate] • Target: Fundamental Counting and Linear Permutations

Find the number of distinct arrangements of the letters of the word STATISTICS.

Final Answer: 50400. Applying Permutations & Combinations principles, $\frac{10!}{3!\,3!\,2!} = 50400$ Therefore, the verified correct option is 50400.

13 [Intermediate] • Target: Permutations with Repetitions and Identical Items

In how many distinct ways can 11 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 3628800. Applying Permutations & Combinations principles, $(n-1)! = (11-1)! = 3628800$ Therefore, the verified correct option is 3628800.

14 [Intermediate] • Target: Combinations and Committee Selections

In how many distinct ways can 10 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 362880. Applying Permutations & Combinations principles, $(n-1)! = (10-1)! = 362880$ Therefore, the verified correct option is 362880.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the number of distinct arrangements of the letters of the word ARRANGE.

Final Answer: 1260. Applying Permutations & Combinations principles, $\frac{7!}{2!\,2!} = 1260$ Therefore, the verified correct option is 1260.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

In how many distinct ways can 9 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 40320. Applying Permutations & Combinations principles, $(n-1)! = (9-1)! = 40320$ Therefore, the verified correct option is 40320.

17 [Challenging] • Target: Fundamental Counting and Linear Permutations • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the number of distinct arrangements of the letters of the word BALLOON.

Final Answer: 1260. Applying Permutations & Combinations principles, $\frac{7!}{2!\,2!} = 1260$ Therefore, the verified correct option is 1260.

18 [Challenging] • Target: Permutations with Repetitions and Identical Items • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the number of distinct arrangements of the letters of the word ADDRESS.

Final Answer: 1260. Applying Permutations & Combinations principles, $\frac{7!}{2!\,2!} = 1260$ Therefore, the verified correct option is 1260.

19 [Challenging] • Target: Combinations and Committee Selections • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In how many distinct ways can 8 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 5040. Applying Permutations & Combinations principles, $(n-1)! = (8-1)! = 5040$ Therefore, the verified correct option is 5040.

20 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Final Answer: 120. Applying Permutations & Combinations principles, $(n-1)! = (6-1)! = 120$ Therefore, the verified correct option is 120.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{5}P_{2} \).

a. 25

b. 21

c. 10

d. **20** ✓

\({}^{n}P_{r} = \frac{n!}{(n-r)!} \implies {}^{5}P_{2} = 5 \times 4 = 20 \).

22 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{6}P_{2} \).

a. 12

b. 15

c. 36

d. **30** ✓

\({}^{n}P_{r} = \frac{n!}{(n-r)!} \implies {}^{6}P_{2} = 6 \times 5 = 30 \).

23 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{7}P_{2} \).

a. 21

b. **42** ✓

c. 49

d. 14

\({}^{n}P_{r} = \frac{n!}{(n-r)!} \implies {}^{7}P_{2} = 7 \times 6 = 42 \).

24 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{8}P_{2} \).

a. 64

b. **56** ✓

c. 16

d. 28

\({}^{n}P_{r} = \frac{n!}{(n-r)!} \implies {}^{8}P_{2} = 8 \times 7 = 56 \).

25 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \({}^{9}P_{2} \).

a. **72** ✓

b. 81

c. 18

d. 36

\({}^{n}P_{r} = \frac{n!}{(n-r)!} \implies {}^{9}P_{2} = 9 \times 8 = 72 \).

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations $ {}^{10}P_{2} $.

a. **90** ✓ **b.** 100

c. 20 **d.** 45

$ {}^{n}P_{r} = \frac{n!}{(n-r)!} \implies {}^{10}P_{2} = 10 \times 9 = 90 $.

27 [Foundational] • Target: Permutations & Combinations Practice Exercises

A committee of 4 people is to be chosen from $ 5 $ men and $ 4 $ women. How many committees consist of exactly 2 men and 2 women?

a. **60** ✓ **b.** 120

c. 55 **d.** 36

$ \binom{5}{2} \times \binom{4}{2} = \frac{5\times4}{2} \times \frac{4\times3}{2} = 60 $.

28 [Foundational] • Target: Permutations & Combinations Practice Exercises

A committee of 4 people is to be chosen from $ 6 $ men and $ 5 $ women. How many committees consist of exactly 2 men and 2 women?

a. 44 **b.** 145

c. 300 **d.** **150** ✓

$ \binom{6}{2} \times \binom{5}{2} = \frac{6\times5}{2} \times \frac{5\times4}{2} = 150 $.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Fundamental Counting and Linear Permutations

In how many ways can 5 different books be arranged on a shelf?

a. 120 ✓ **b.** 24

c. 60 **d.** 720

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Fundamental Counting and Linear Permutations

Evaluate the number of permutations of 7 distinct items taken 3 at a time, ${}^{7}P_{3}$.

a. 210 ✓

b. 35

c. 42

d. 840

${}^{7}P_{3} = \frac{7!}{(7-3)!} = 7 \times 6 \times 5 = 210$.

31 [Intermediate] • Target: Fundamental Counting and Linear Permutations

How many 4-digit even numbers can be formed using digits $\{1, 2, 3, 5, 7, 8\}$ without repetition?

a. 120 ✓

b. 60

c. 240

d. 180

Last digit must be 2 or 8 (2 choices). First three digits from remaining 5 digits: ${}^{5}P_{3} = 60$. Total $= 60 \times 2 = 120$.

32 [Intermediate] • Target: Permutations with Repetitions and Identical Items

Find the number of distinct permutations of all the letters in the word PARALLEL.

a. 3360 ✓

b. 6720

c. 1680

d. 40320

8 letters total: 3 L's, 2 A's, 1 P, 1 R, 1 E. Total $= \frac{8!}{3!\,2!} = \frac{40320}{12} = 3360$.

33 [Intermediate] • Target: Permutations with Repetitions and Identical Items

How many 6-letter arrangements of PEPPER begin with the letter P?

a. 30 ✓

b. 60

c. 20

d. 120

Fix one P first. Remaining 5 letters: 2 P's, 2 E's, 1 R. Arrangements $= \frac{5!}{2!\,2!} = \frac{120}{4} = 30$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Permutations with Repetitions and Identical Items

How many 5-digit codes can be formed from digits $\{3, 3, 3, 7, 7\}$?

a. 10 ✓

b. 20

c. 5

d. 120

$\frac{5!}{3!\,2!} = \frac{120}{6 \times 2} = 10.$

35 [Intermediate] • Target: Combinations and Committee Selections

A committee of 4 is to be chosen from 9 people. How many distinct committees can be formed?

a. 126 ✓

b. 3024

c. 84

d. 252

${}^{9}C_{4} = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126.$

36 [Intermediate] • Target: Combinations and Committee Selections

A team of 5 is selected from 6 boys and 4 girls. How many teams contain exactly 3 boys and 2 girls?

a. 120 ✓

b. 60

c. 210

d. 100

Choose boys: $\binom{6}{3} = 20$. Choose girls: $\binom{4}{2} = 6$. Total $= 20 \times 6 = 120.$

37 [Intermediate] • Target: Combinations and Committee Selections

From 10 points on a plane where no three are collinear, how many distinct triangles can be drawn?

a. 120 ✓

b. 720

c. 90

d. 45

${}^{10}C_{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In how many ways can 4 boys and 3 girls sit in a row if all 3 girls must sit together?

a. 720 ✓

b. 5040

c. 144

d. 288

Treat 3 girls as 1 block. Total units: 4 boys + 1 block = 5 units ($5! = 120$). Girls within block: $3! = 6$. Total $= 120 \times 6 = 720$.

39 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In how many ways can 5 people sit around a circular dinner table?

a. 24 ✓

b. 120

c. 60

d. 12

Circular permutations: $(n-1)! = (5-1)! = 4! = 24$.

40 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A committee of 4 is selected from 5 men and 5 women. How many committees contain at least one woman?

a. 205 ✓

b. 210

c. 195

d. 150

Total selections $= \binom{10}{4} = 210$. All-men selections $= \binom{5}{4} = 5$. At least one woman $= 210 - 5 = 205$.

41 [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

In how many distinct ways can 7 different people be seated around a circular table (rotations considered identical)?

a. **720** ✓

b. 766

c. 647

d. 611

Applying Permutations & Combinations principles, $(n-1)! = (7-1)! = 720$ Therefore, the verified correct option is 720.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Fundamental Counting and Linear Permutations • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the number of distinct arrangements of the letters of the word GIGGLE.

a. 214

b. 141

c. **180** ✓

d. 207

Applying Permutations & Combinations principles, $\frac{6!}{3!} = 180$ Therefore, the verified correct option is 180.

43 [Challenging] • Target: Permutations with Repetitions and Identical Items • Cross-Lesson: 8.01 Integration and Kinematics

Find the number of distinct arrangements of the letters of the word LEVEL.

a. 12

b. 29

c. **30** ✓

d. 76

Applying Permutations & Combinations principles, $\frac{5!}{2!\,2!} = 30$ Therefore, the verified correct option is 30.

44 [Challenging] • Target: Combinations and Committee Selections • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the number of distinct arrangements of the letters of the word PEPPER.

a. 33

b. 17

c. 34

d. **60** ✓

Applying Permutations & Combinations principles, $\frac{6!}{3!\,2!} = 60$ Therefore, the verified correct option is 60.

45 [Challenging] • Target: Restricted Arrangements and Separations • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the number of distinct arrangements of the letters of the word BANANA.

a. 52

b. 71

c. **60** ✓

d. 62

Applying Permutations & Combinations principles, $\frac{6!}{3!\,2!} = 60$ Therefore, the verified correct option is 60.

© ClipSAT — clipsat.org | All Rights Reserved